VIDEO DEEPFAKE DETECTION

Explainable AI Forensic Analysis Report

ANALYSIS ID
sample_000_003...
2026-11-05T18:14:13Z

FORENSIC VERDICT	MANIPULATION PROBABILITY	THRESHOLDS	CLIPS / DURATION
LIKELY MANIPULATED	**93.2%** P(REAL): 6.8%	Authentic: <0.30 Manipulated: ≥0.70	**60 clips** (15.84s) Time: 0.96s

Forensic Summary & Evidence Synthesis

Forensic analysis classified this video as LIKELY MANIPULATED with P(FAKE) = 93.2% across 60 evaluated clips. The spatial branch drove 50.4% of the decision, revealing prominent synthetic boundaries and textural anomalies concentrated around the lips (+0.18), skin (+0.62), eyes (+1.04), nose (+0.44). The temporal branch accounted for 49.6%, indicating frame-to-frame inconsistencies. The evidence consistently exceeds the configured manipulation threshold.

Branch Attribution (Exact Shapley Decomposition)

Branch	Diagnostic Focus	Evidence Share (%)	Shapley Value (pl
Spatial Head (S)	Appearance, facial anatomy, synthesis boundary artifacts	50.4%	+0.2175
Temporal Head (T)	Motion continuity, temporal jitter, inter-frame coherence	49.6%	+0.2142

Facial Component Attribution (FCG Query Space)

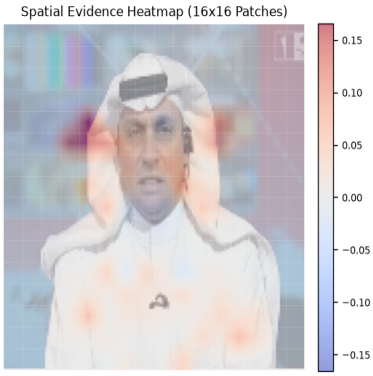
Facial Component	Signed Logit Contribution	Normalized Share (%)	Evidence Pushes Towards	Importance
Lips	+0.1833	8.1%	FAKE	LOW
Skin	+0.6171	27.1%	FAKE	MEDIUM
Eyes	+1.0386	45.6%	FAKE	HIGH
Nose	+0.4383	19.2%	FAKE	MEDIUM

Visual Forensic Heatmaps (16x16 Patch Decomposition)

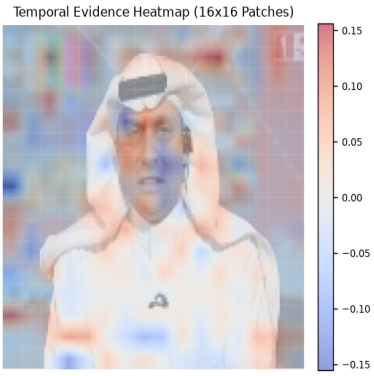
Extracted Face Crop



Spatial 16x16 Heatmap



Temporal 16x16 Heatmap



Video Quality & Robustness Assessment

Resolution: 640x480 | **FPS:** 25.0 | **Sharpness:** Crisp (206.92 var) | **Compression:** Moderate

- High image sharpness (Laplacian variance 206.9). Fine facial textures preserved.
- Balanced lighting (mean luminance 158.1/255).
- Standard definition video (640x480).

Methodology & Architecture: Yue-Hua Han et al., 'Towards More General Video-based Deepfake Detection through Facial Component Guided Adaptation for Foundation Model' (CVPR 2025, arXiv:2404.05583). Attributions computed via exact Shapley decomposition and additive LayerNorm-Linear projections. Formal video deepfake forensic analysis report.